main.py

Visualize | Run ▶

```python
import numpy as np

# Step 1: Create a one-dimensional array
arr = np.arange(1, 11)

print("Original Array:", arr)

# Step 2: Perform slicing operations
print("First five elements:", arr[:5])
print("Last five elements:", arr[5:])
print("Elements from index 2 to 6:", arr[2:7])
print("Every second element:", arr[::2])

# Step 3: Calculate statistical measures
print("Sum:", np.sum(arr))
print("Mean:", np.mean(arr))
print("Maximum:", np.max(arr))
print("Minimum:", np.min(arr))

# Step 4: Apply broadcasting
```

Output

```
Original Array: [ 1  2  3  4  5  6  7  8  9 10]
First five elements: [1 2 3 4 5]
Last five elements: [ 6  7  8  9 10]
Elements from index 2 to 6: [3 4 5 6 7]
Every second element: [1 3 5 7 9]
Sum: 55
Mean: 5.5
Maximum: 10
Minimum: 1
Array after adding 5: [ 6  7  8  9 10 11 12 13 14 15]
```